A2Z MIS 360

Banking Management Intelligence — Audit-Locked

Version	v8.14
Audit gates	109 (G104-G109 defense-in-depth perimeter)
System stocks	6 (6/6 wired)
Feedback loops	15 (15/15 wired)
Learning loops	3 (Meadows' highest-value type)
Build	2026-05-01T17:51:12

To verify any claim in this document, run `python scripts/audit.py`

Foreword

Most banking platforms have a documentation problem: the slide decks the sales team uses describe one product; the codebase implements another. The drift compounds. Six months in, no one trusts either source.

A2Z does not have that problem. This document was rendered from the same registries that the 109-gate audit suite verifies on every build. Numbers in this document trace to registry paths; if a future build changes the registry, the next regeneration will reflect the change. To verify the platform's current state, run the command on the cover.

Every part ends with a section titled **What this document does not claim** — features that are roadmap rather than shipped, integrations that are designed but not deployed, outcomes that are projections rather than measurements. This is unusual for sales collateral. It is what makes the result trustworthy.

— A2Z Platform Engineering

PART 1 — Platform Overview

The strategy-execution gap

Banks struggle to translate strategic goals into individual actions. Quarterly board targets do not flow to teller-level KPIs. Manual cascade in Excel breaks down. Staff receive performance feedback monthly — too late to change behaviour mid-cycle. Audit trails are scattered across 10-15 systems; CBK reviews require manual reconstruction.

The A2Z approach

A2Z is a banking management intelligence platform. Three discipline patterns differentiate it:

Audit-locked invariants. The codebase enforces its own architectural rules through 6 defense-in-depth gates (G104-G109). Engineering decisions documented in the charter become permanent invariants — future regressions fail the build.

Systems-layer first. Stocks, flows, invariants, and composites are first-class objects in the codebase. A2Z has 6 stocks (6/6 wired), 15 feedback loops (15/15 wired), of which 3 are Meadows' highest-value learning loops.

Honest scope. Every artifact ends with what it does NOT claim. Roadmap items are marked. ROI projections are distinguished from measurements.

PART 2 — Architecture

Six tiers, audit-locked

The platform organizes into six tiers. Tiers 1-5 are existing; tier 6 is the Living Documentation System (this artifact).

Tier 1 — Domain engines

120 modules in utils/ — pure deterministic logic. Zero presentation. Each engine implements one banking standard.

Tier 2 — Systems-layer registries

6 system stocks (system_stocks.py) + 15 feedback loops (system_flows.py) + invariants (system_invariants.py) + composite_scores.py. Per Meadows: a system is its feedback loops.

Tier 3 — Audit perimeter

scripts/audit.py with 109 gates. Six form the defense-in-depth perimeter (G104-G109). The build fails if any gate fails.

Tier 4 — Technical-grade documentation

A2Z_SYSTEMS_CHARTER.md (297 lines, 14 sections) + A2Z_V7_RETROSPECTIVE.md (282 lines) + A2Z_V8_RETROSPECTIVE.md (364 lines) + 84 CHANGELOG files.

Tier 5 — UI surfaces

4 dedicated engine pages (88-91) + 16 enhanced pages + admin operations. Page 91 systems view shows mode + circuit + latency in one place.

Tier 6 — Living Documentation

This artifact. Sales-grade rendering with audit-locked claim validation.

PART 3 — Systems Layer

System stocks

A2Z tracks 6 system stocks. A stock is a quantity that accumulates over time (loan portfolio, deposit base, capital, etc). Each is registered in utils/system_stocks.py with explicit status + data_source provenance (locked by audit gate G107).

Stock ID	Name	Status	Data source
customer_base	Customer base	WIRED	—
loan_portfolio	Loan portfolio (gross outstandin	WIRED	—
deposit_base	Deposit base (customer deposits)	WIRED	—
npl_inventory	NPL inventory (non-performing lo	WIRED	—
dormant_accounts	Dormant accounts	WIRED	—
capital_base	Capital base (Tier 1 + Tier 2)	WIRED	—

Feedback loops

A2Z has 15 designed feedback loops, all 15/15 wired in code. Three are Meadows' highest-value learning loops where outcomes recalibrate behaviour. Each loop has a from-engine, to-engine, payload, and DDD pattern (Published Language, Customer-Supplier, Anti-Corruption Layer, Conformist, Open Host Service, or Shared Kernel).

Loop	From → To	Pattern	Status
L01	Credit Risk → Credit Risk	Published Language	✓ WIRED
L02	Profitability → Strategy & Cas	Published Language	✓ WIRED
L03	Smart Alerts & → Performance Me	Open Host Service	✓ WIRED
L04	Cross-sell & N → Operational Ri	Published Language	✓ WIRED
L05	Branch & Chann → Customer Intel	Published Language	✓ WIRED
L06	Daily-Risk Tri → Treasury & ALM	Published Language	✓ WIRED
L07	Compliance / A → Compliance / A	Published Language	✓ WIRED
L08	HR Intelligenc → HR Intelligenc	Published Language	✓ WIRED
...	(7 more)	—	—

PART 4 — Resilience + Observability

Per CBK Operations Resilience Guidelines (2019), live FLEXCUBE calls must survive transient failures without cascading impact. A2Z's resilience layer is locked by audit gate G108.

Resilience layers

Retry. 3 attempts with exponential backoff (1s/3s/9s) + ±20% jitter (v8.1 + v8.8)

Circuit breaker. Trips OPEN after 5 consecutive failures; 60s open duration; half-open probe pattern (v8.1)

Latency telemetry. Per-endpoint p50/p95/p99 over rolling 200-sample window; circuit-open suppression (v8.2)

Restart-free admin. reset_circuit() + replay_events() with audit trails (v8.9)

Observability triangle

Three operator-facing surfaces visible from page 91 systems view, answering three different questions:

Surface	Question	Values	Shipped
Mode banner	Which path are we on?	synthetic / mock / live	v7.10
Circuit banner	Is the path healthy?	closed / intermittent / OPEN	v8.1
Latency expander	How fast is the path?	p50 / p95 / p99 per endpoint	v8.2

PART 5 — Audit Perimeter

The audit suite has 109 gates total. Six form the defense-in-depth perimeter that locks the v7.x → v8.x architecture as permanent invariants. Adding a regression-prone feature requires either passing every gate or extending the audit.

Gate	What it locks	Shipped
G104	Engine migration ratchet	v7.0.1
G105	Strict invariant registry usage	v7.1
G106	Loop round-trip-testability	v7.15
G107	Stock data_source provenance	v7.15
G108	FLEXCUBE resilience + observability	v8.3
G109	PUBLISHED_LANGUAGE payload_version	v8.7

Verification: the audit suite runs in seconds. Run `python scripts/audit.py` on any version of the platform; any failure is a defect.

PART 6 — Regulatory Alignment

A2Z aligns with the following regulatory frames at the architecture / engine level. Per-bank deployment review is required for production certification.

- ✓ CBK Operations Resilience Guidelines (2019)
- ✓ CBK Prudential Guidelines
- ✓ Data Protection Act 2019
- ✓ IFRS 9 / IFRS 7 / IFRS 16 / IFRS 15
- ✓ Basel III

Honest scope: A2Z does not currently hold SOC 2 Type II or ISO 27001 certifications. References to readiness reflect architectural alignment with the standards' control families, not awarded certifications.

PART 7 — Canonical References

A2Z's discipline pattern derives from a small set of canonical references. The same sources cited in our retrospectives.

- Donella Meadows, *Thinking in Systems* (2008)
- Eric Evans, *Domain-Driven Design* (2003)
- Michael Nygard, *Release It!* (2007)
- Sam Newman, *Building Microservices* (2015)
- CBK Operations Resilience Guidelines (2019)

Internal canonical references (the campaign canon):

- A2Z_SYSTEMS_CHARTER.md (297 lines)
- A2Z_V7_RETROSPECTIVE.md (282 lines)
- A2Z_V8_RETROSPECTIVE.md (364 lines)
- A2Z_LIVING_DOCS_PLAN.md (588 lines)

PART 8 — What this document does not claim

This artifact is rendered from registries that the codebase audits on every build. Numbers that appear here trace to a registry path; if a future build changes the registry, the next regeneration of this artifact will reflect the change. To verify the platform's current state, run python scripts/audit.py. The list below enumerates what this artifact deliberately does NOT claim — features that are roadmap rather than shipped, integrations that are designed but not deployed, outcomes that are projections rather than measurements.

Roadmap items (not yet shipped)

→ core_banking/temenos_t24: roadmap

→ credit_bureaus/transunion: roadmap

→ mobile_money/airtel_money: roadmap

→ payment_networks/swift: roadmap

→ payment_networks/kepss: roadmap

→ hris/workday: roadmap

→ hris/sap_successfactors: roadmap

→ crm/salesforce: roadmap

→ crm/dynamics_365: roadmap

→ gl_erp/sap_finance: roadmap

→ gl_erp/oracle_financials: roadmap

Honest scope statements

- Encryption at-rest and in-transit are deployment-specific patterns; A2Z relies on host infrastructure (cloud/on-prem) for actual implementation.
- MFA is designed but not yet wired into the default login flow.
- SAML/OIDC SSO is roadmap, not shipped.
- HSM key-management integration is bank-deployment-specific; A2Z provides hooks but not bundled HSM.
- SOC 2 and ISO 27001 are roadmap goals, not awarded certifications.
- Multi-process state (circuit breaker + latency telemetry shared across gunicorn workers / kubernetes pods) needs Redis backend; v8.x ships single-process pattern, multi-process is v9.x roadmap.
- Most external system integrations (Salesforce, Workday, SAP, Temenos) are roadmap. v8.x deployments rely on the local A2Z Blueprint conventions.
- FLEXCUBE 12 is the only fully-implemented core-banking adapter as of v8.x. Other cores require new ACL implementations following the same pattern.
- M-PESA + CRB Kenya integrations are designed at engine level but require API credentials for production deployment; not pre-bundled.
- ISO 20022 messaging is roadmap, not shipped.

- Real-time payment-network connectivity (RTGS, ACH) is roadmap.
- A2Z does not currently have any signed production-reference deployments with measurable outcomes.
- Numbers like '99.2% repayment' or '$25-30B opportunity' come from external companies' published research, not from A2Z deployments. They appear only in 'Cited industry research' sections.
- Bank logos and named-customer testimonials require signed reference agreements; the magazine-generator must check this flag before rendering.
- The Ecobank Kenya design-partner relationship is for product development; it is not represented as a customer success story in published collateral.
- Future production deployments with measurable outcomes will be added here as 'a2z_reference_deployments' entries with may_appear_in_collateral: true.
- A2Z has no measured ROI outcomes from production deployments. Any number framed as 'achievable ROI' is a projection, not a measurement.
- Pricing model (per-bank licence, per-user, per-transaction) is not yet defined and must NOT appear in collateral as a fixed figure.
- The 'numbers like 23% → 85%' from the original draft are removed because they have no methodology source.
- ROI commitments to prospects must be made with explicit pre/post measurement methodology + bank-specific baselines, not generic claims.

Sales conversations grounded in this discipline produce more trust, not less. The 36+ consecutive clean-first-try batches that built A2Z were enabled by exactly this honesty.